01 / CURVED PATH STUDY

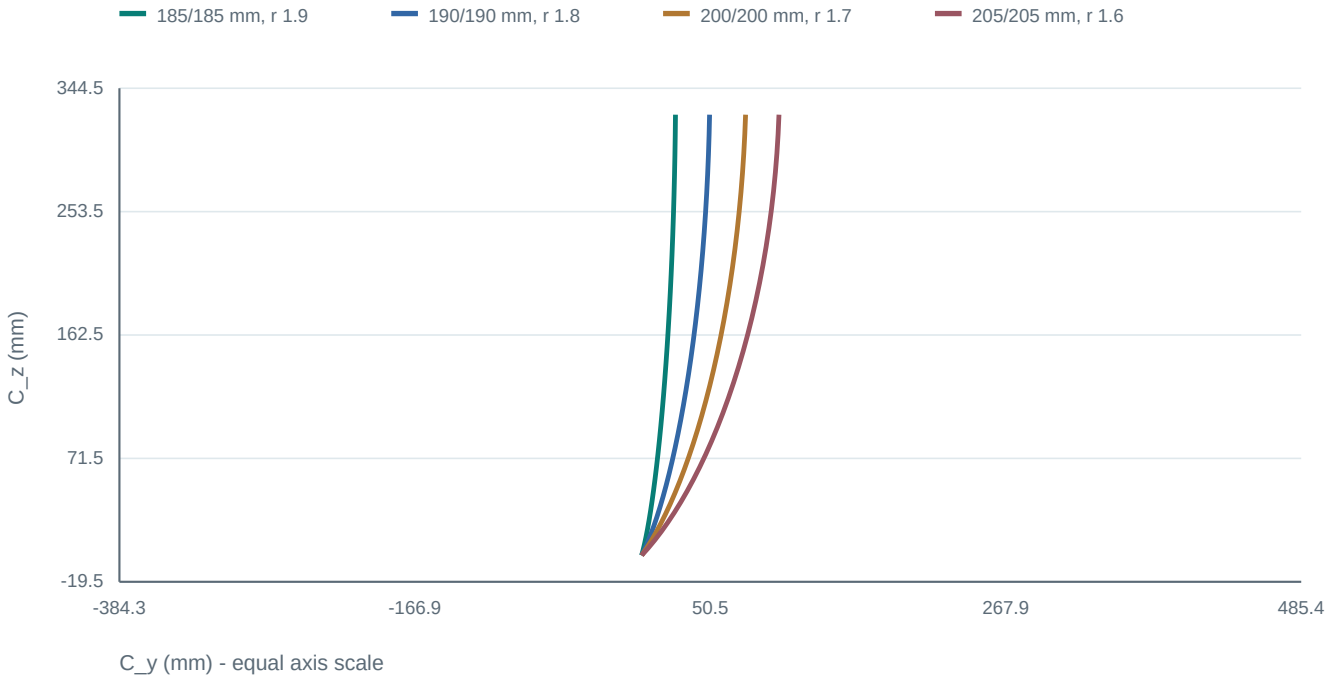
Curved paths from ratio changes

325 mm total vertical travel, including two assumed 20 mm reserves; 285 mm useful.

This study searches useful curved wheel-centre paths with 25, 50, 75 or 100 mm of sideways excursion. All leading examples retain **alpha=0 and original assembly phase phi0=0**. Changing the guide ratio produces the curve; no mounting tilt or indexing is introduced.

Target mm	Equal links mm	Guide ratio	Actual lateral mm	Hip peak Nm
25	185	1.9	24.89	38.39
50	190	1.8	49.98	37.36
75	200	1.7	76.48	37.09
100	205	1.6	101.05	35.89

Normal strokes aligned at their extended start; equal physical axis scales



Balanced geometric seed: 190/190 mm with guide 9:5. It provides 49.98 mm lateral excursion and 325/285 mm total/useful height. These compact examples fail the assumed normal 220 mm coplanar wheel/hip exclusion. Larger clearance-filtered alternatives are shown separately; actual physical packaging remains unresolved.

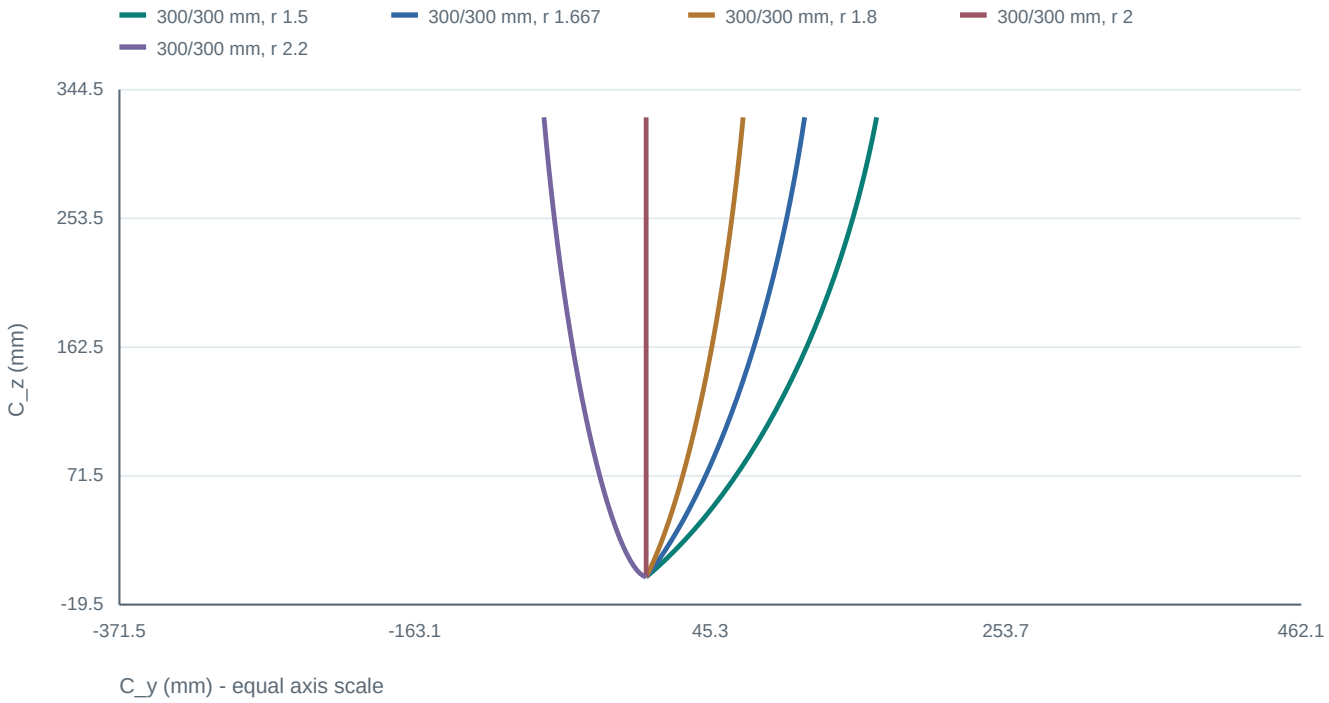
02 / CURVED PATH STUDY

The ratio bends the path

Same 300/300 mm links, same q start=100 degrees, same 325 mm vertical stroke.

With t=q-pi and k=r-1, the original phase-zero model is `x=L2 cos(t)-L3 cos(k t); z=L2 sin(t)+L3 sin(k t)`. Equal links at r=2 move vertically. Reducing r below 2 allows sideways motion as the leg compresses; increasing r above 2 bends the normal path the other way.

Ratio sensitivity isolated from changes in link length



Guide r	Normal sideways mm	Useful hip ROM deg	45 kg hip peak Nm
1.5	162.61	49.97	46.48
1.667	111.81	46.77	50.40
1.8	68.42	45.15	53.48
2	0.00	43.62	58.22
2.2	72.08	41.42	63.61

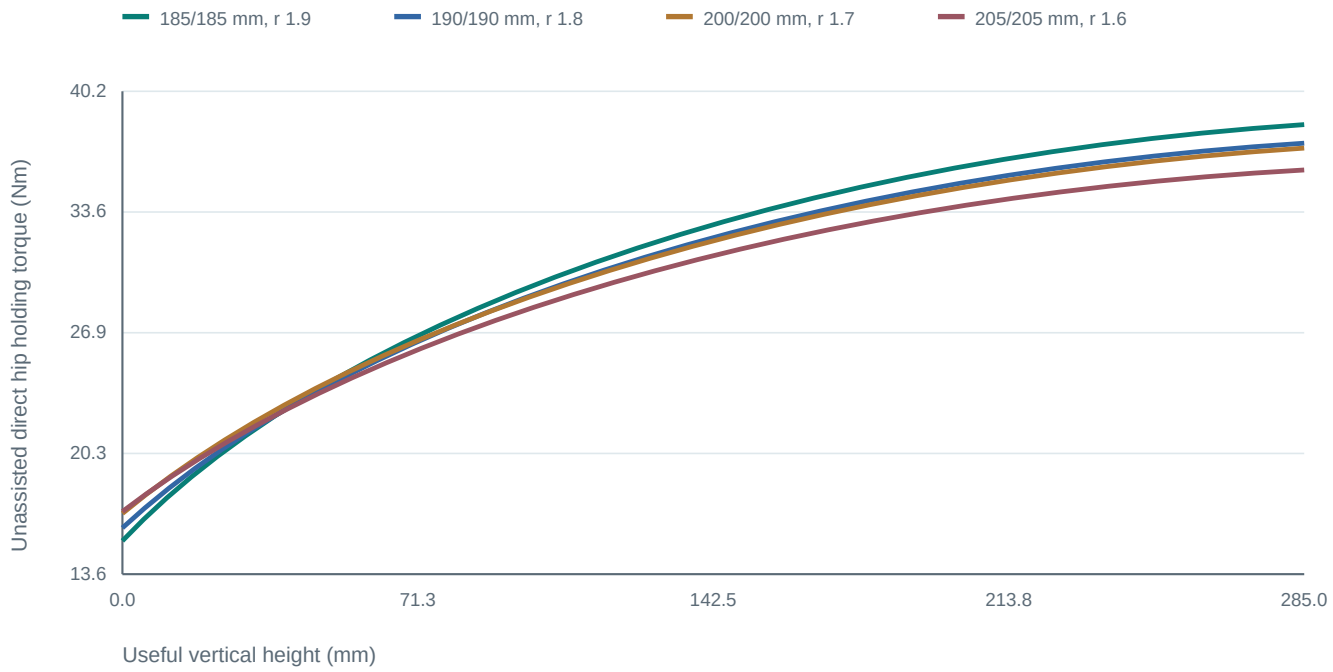
The guide ratio is ra/rb. With rb=60 mm, the four compact examples have ra=114/108/102/96 mm. These are pitch-radius relations, not selected pulley tooth counts, belt pitches or strength ratings. The 2:1 zero-excursion case is a baseline, not a curved-path winner.

03 / CURVED PATH STUDY

Force, speed and sideways loading

Pure vertical equivalent load: 45 kg robot /4 corners =110.3625 N per leg; massless moving links.

Individual direct-hip holding torque curves over useful travel



L mm /r	Peak /RMS Nm	Hip ROM deg	Speed rad/s	Side-load peak Nm
185 /1.9	38.39 /31.72	61.08	1.785	38.56
190 /1.8	37.36 /31.08	61.82	1.706	37.71
200 /1.7	37.09 /30.97	61.65	1.626	37.66
205 /1.6	35.89 /30.18	62.98	1.616	36.54

Speed assumes 0.25 m/s constant vertical movement. Side-load peak is a robust envelope for either sign of Fx with |Fx/Fz|<=0.3, an explicitly assumed sensitivity. `|tau_hip|=|Fx Jx+Fz Jz|`. A curve changes both force projection and required joint speed; it does not provide free energy.

For the relative knee coordinate, **tau_knee=-tau_hip/r** and **knee speed=-r hip speed**. The factor one-half applies only at r=2. Any motor gearbox is a separate mapping. Height-RMS corresponds to constant vertical speed; real thermal duty needs time weighting. Ideal useful lift work is 31.453 J per leg in every case. 3/4g are held-load multiples, not impact predictions or motor ratings.

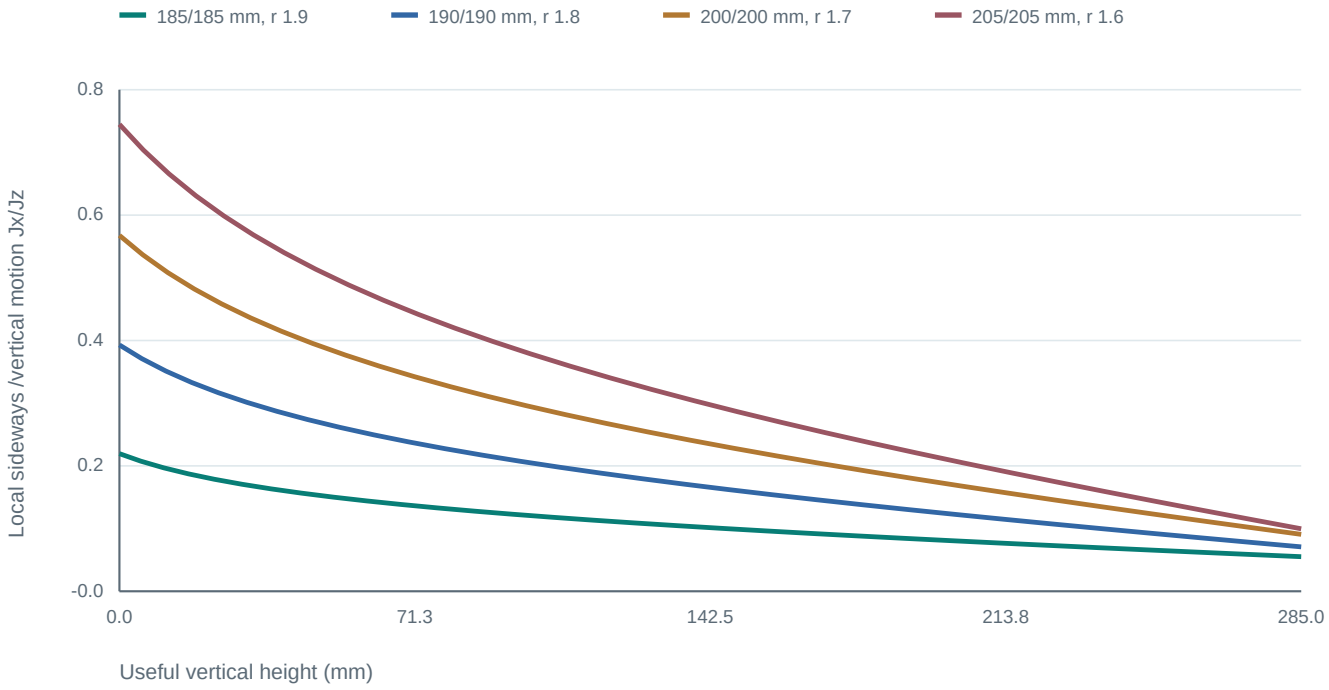
04 / CURVED PATH STUDY

Allowed movement is not stiffness

A single guided joint permits coupled horizontal and vertical motion along the local tangent.

The slope Jx/Jz measures horizontal movement per unit vertical movement. Larger lateral excursion allows the wheel centre to move sideways as the leg compresses, but does not establish a softer suspension. A locked actuator can remain stiff despite a curved path.

Horizontal-to-vertical movement ratio through each useful stroke




```
J=[Jx,Jz] (metres/radian)
C = J J^T / K_eff
Cxx=Jx^2/K_eff; Cxz=Jx Jz/K_eff; Czz=Jz^2/K_eff
```

This local one-DOF compliance matrix applies only for a stable effective joint stiffness K_eff in Nm/rad, with the permitted coupled motion free to occur. It has rank one. K_eff must include spring/controller stiffness and load-dependent geometric stiffness. Without those inputs, an absolute horizontal stiffness or spring constant would be invented.

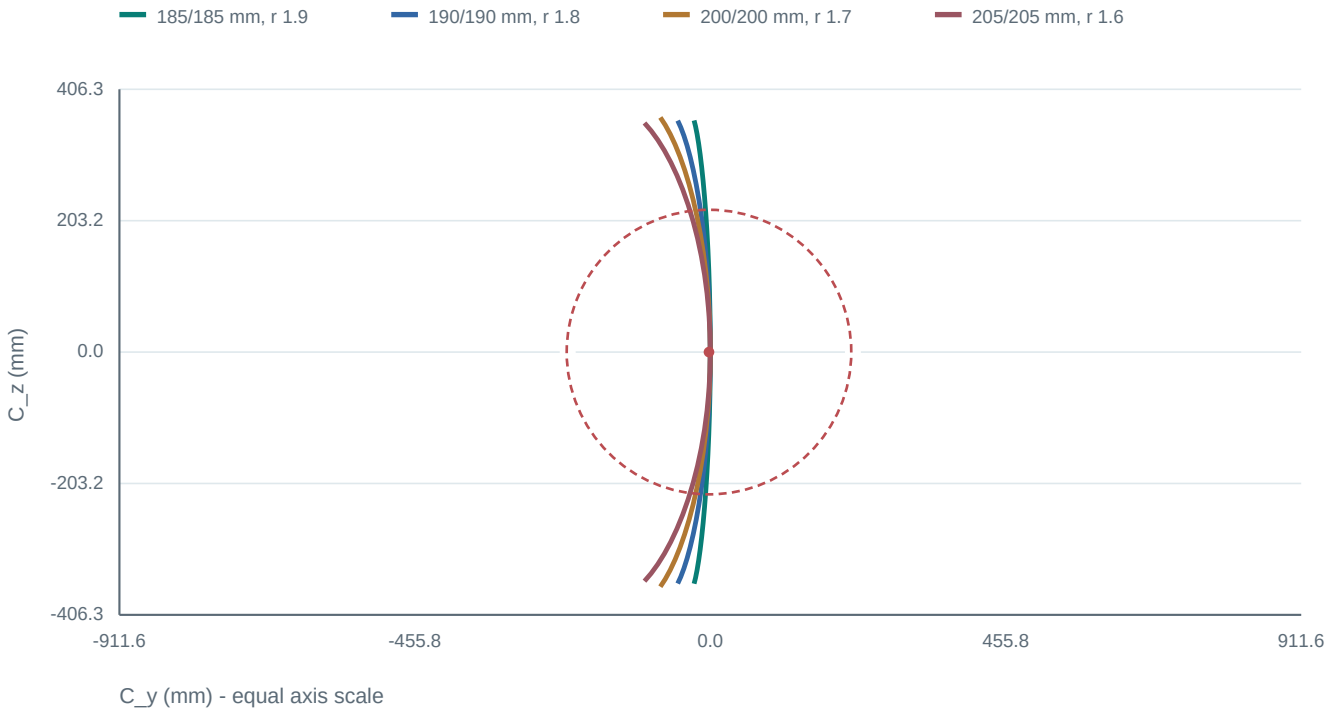
If vertical motion is blocked by rigid contact or another constraint, the coupled sideways motion can also be blocked. Ground contact, traction, tire deformation and wheel rotation decide whether the path is helpful compliance, rolling movement or scrub. Existing wheel-spin cancellation is not a rolling constraint. At the phase-zero fold Jx=0, so the permitted sideways motion vanishes locally even though vertical inversion remains regular.

05 / CURVED PATH STUDY

Inversion and wheel packaging

Full ideal envelope q=100-260 degrees; a 325 mm normal working stroke is only part of it.

Full paths and the assumed 220 mm projected wheel-centre exclusion around hip A



Target mm	Compact L mm	Normal hip distance mm	220-filter L mm	220-filter ratio
25	185	33.14	280	1.93
50	190	32.89	285	1.85
75	200	37.77	295	1.775
100	205	29.23	300	1.705

Strict phase-zero symmetry gives x(-t)=x(t), z(-t)=-z(t). The full path moves sideways toward the fold and returns on the other branch, so **full endpoint horizontal drift is zero despite a substantial lateral excursion**. Lateral excursion, signed drift and turning points are therefore reported separately.

All compact examples pass the ideal height-inverse check through inversion, but every near-equal coplanar full sweep fails clearance: at the fold |C|=|L2-L3|. The larger 280/285/295/300 mm alternatives clear the normal branch only. Measured axial wheel/chassis and belt geometry is still required. Historical dual-rope remains **IMPOSSIBLE /rejected**.

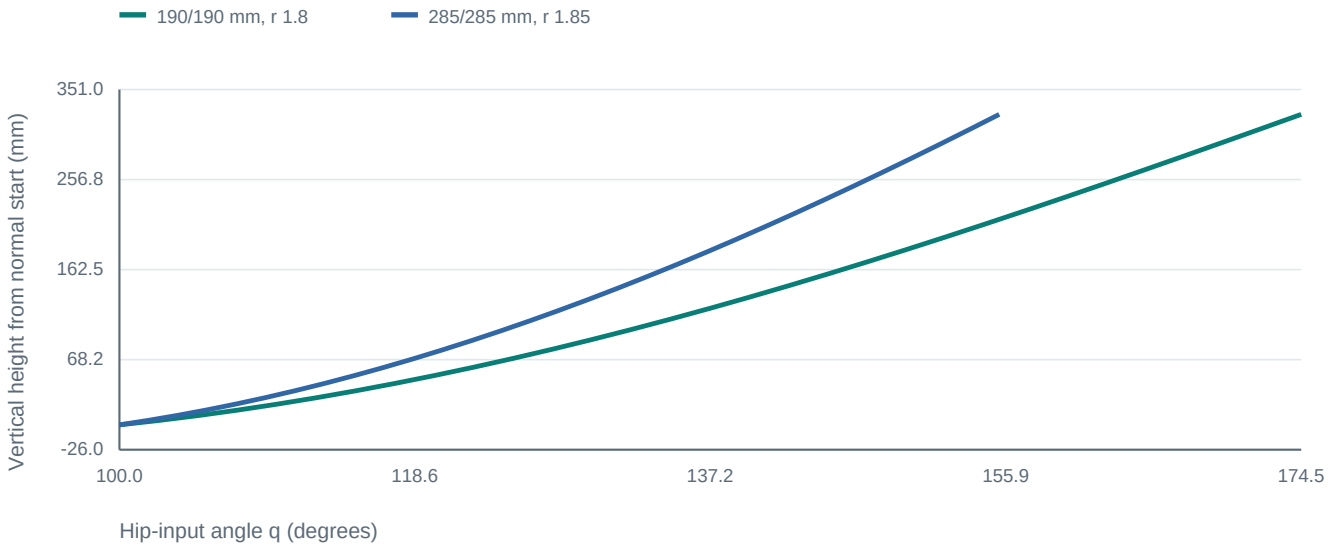
06 / CURVED PATH STUDY

Search priorities and effective ROM

2,953,002 windows enumerated; hard constraints first, separate force and size leaders.

Variable /filter	Policy
Links /guide	L3 150-400 step5 mm; L2/L3 .85-1.15 step.025; r1.3-2.6 step.005 plus exact5/3
Angle /height	q start100-140 step2.5 deg; end solved for325 mm, <=175; full100-260
Path target	Normal lateral excursion25/50/75/100 +/-2.5 mm; monotone sideways normal motion
Feasibility	Jz>=5 mm/rad across full envelope; L2>60(r+1) pitch separation; 5 deg fold guard
Scenarios /ranking	Equal or within5%; geometry-only vs normal220; minimize peakJz or link sum separately
Verification	33 coarse poses; retained continuous derivative extrema +2001-pose force/integrals

The same height reserves trim different angular intervals



Effective ROM solves height roots: qlow=z^-1(zstart+20), qhigh=z^-1(zend-20). The 190 mm, 9:5 case has total q=100-174.484 deg and useful q=109.294-171.110 deg, giving 61.816 deg hip ROM. The 100/175 degree limits are study assumptions, not hardware stops. Separate 90/95 degree start sensitivities expand the full envelope consistently and can permit still shorter links.

No weighted blended score is used. Force leaders sometimes sit near a lateral tolerance edge; rounded-ratio examples are chosen for understandable comparisons. Finite grid and pool retention do not prove global or manufacturing optima.

07 / CURVED PATH STUDY

Validation and next design choice

Independent original models, numerical tests and actual constrained Pymunk geometry.

10,788 scalar comparisons against original Jonathan V2 geometry/statics and Andy 2D statics pass; maximum discrepancy 2.274e-13 mm/Nm. Six new tests plus five force and five inversion tests pass. Independent review checks the new excursion, inverse, force and compliance interpretation.

Actual Pymunk case	2 ms path error mm	1 ms mm	0.5 ms mm
185/185 mm, r 1.9	0.07369	0.03659	0.01823
190/190 mm, r 1.8	0.06784	0.03368	0.01678
205/205 mm, r 1.6	0.06899	0.03425	0.01707

All three fixtures pass at 2/1/0.5 ms and converge. Real motor/pivot/gear constraints generate motion with positive body masses; only initial poses are assigned. Zero gravity/forces and absent collision shapes make this a geometry check, not measured compliance, force, contact or impact validation. Earlier failures remain in their earlier reports.

Start with 190/190 mm and guide 9:5 for the 50 mm case if compact geometry and unresolved axial packaging are acceptable study assumptions. Use the 285 mm, r=1.85 normal-clearance alternative if a coplanar wheel/hip layout is mandatory. Compare the 100 mm case only if more lateral sweep is worth its added rolling/scrub and side-load coupling.

Next specify actual axial/chassis clearances, spring route, controller stiffness, contact conditions and motor inertia/duty before selecting stiffness or motor ratings. Previous spring/motor calculations are separate geometries. No new Motion Lab controls or services are added; existing models, examples/GIFs and rejected decisions stay intact.

Reproduction: curved_path_search.py; test_curved_paths; scripts/verify_curved_geometry.py; scripts/verify_curved_report.cjs --math-only; scripts/build_curved_report.py; then the full browser verifier. Detailed commands, source hashes, decisions and revision map are adjacent. Source base 93fed9f; workspace scope; canonical robot physics NOT APPLICABLE.